

5(a)(i)	<i>any four from</i> 1 synthesise / obtain, therapeutic / correct / normal / dominant, allele / (c)DNA ; 2 from, mRNA / (cells of) healthy person / gene library ; 3 <i>ref. to</i> probe / electrophoresis / sequencing, for identification ; 4 <i>ref. to</i> PCR to amplify, gene / DNA ; 5 <u>restriction</u>, enzyme / endonuclease, + role ; 6 (DNA) <u>ligase</u> + role ; 7 <i>ref. to</i> add promoter ;	4
5(a)(ii)	<i>any two from</i> 1 (can) add / insert, correct / normal / therapeutic / dominant / functional, DNA / allele ; 2 (only) need one, allele / copy, per cell ; 3 to, cure disease / correct phenotype / restore function / restore vision ; 4 to, make / synthesise, correct / functional, protein ; 5 no need to, edit / knock out / remove / delete, faulty allele (as would be the case if faulty allele was dominant) ;	2
5(a)(iii)	<i>any two from</i> 1 LCA patients already, blind / have lost vision / cannot see ; 2 less likely to, harm / add to problem / worsen vision, for LCA patients ; 3 AVP ; e.g. other age-related diseases may involve, many genes / dominant alleles	2
5(b)(i)	<i>any three from</i> 1 wrong base added / wrong bases pair up / base pair mismatch ; 2 with / to, template, DNA / strand(s) ; 3 <i>Taq</i> polymerase, inaccurate / no proofreading ; 4 in, extension / elongation, stage ; 5 mistake / mutation, replicated / copied, (many times) ; 6 AVP ; e.g. high temperature speeds up replication and increases chance of more mistakes	3
5(b)(ii)	<i>any two from</i> 1 AAV / 7m8 / virus / vector, crossed fluid / reached photoreceptors / reached retina ; 2 (virus) delivered (new / <i>GFP</i>) gene / DNA, to photoreceptors / retina cells ; 3 (<i>GFP</i>) gene / DNA, expressed in, photoreceptors / retina cells ; 4 <u>GFP</u> / <u>protein</u> product, is fluorescent ;	2
5(b)(iii)	(7m8) will decrease risk of / increase treatment by, <u>gene therapy</u> ;	1